



National Foresight & Decision Infrastructure
— first module: **Food Security**

"Fill the granary before the drought."

Nash Barara — Product & Design

Chipo — Strategy & Copy

POTRAZ AI4I 2026
DEVELOPMENT TRACK
ENVIRONMENT, CLIMATE & SUSTAINABILITY

THE PROBLEM

The warning arrives too late. From outside.

- Drought and hunger are **rarely a surprise** — the signs build for weeks. The report arrives after the season is lost.
- Warnings come from **global agencies** — country-level, published outside the window for action.
- Even when a district is flagged, **no shared way** to see where limited resources do the most good.

19 of 20 districts projected into high food-security risk by September — without early action.

7× return on timely, well-targeted anticipatory action (WFP).

The hard part is not knowing the drought is coming.
The hard part is **acting early enough, in the right places.**

WHAT HOZI DOES

From early warning to a ready plan.

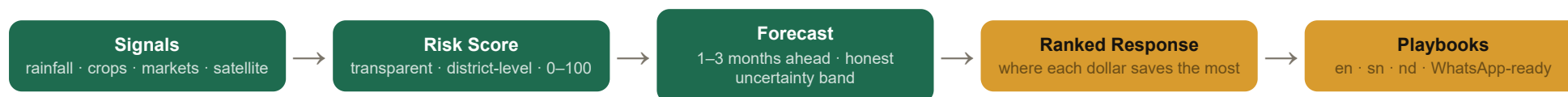
- **District risk map** — ranked watch-list, updated automatically. See where the season is turning.
- **Time-slider forecast** — watch risk spread district by district, July through September.
- **Drill-down** — tap any district: rainfall, drying crops, trajectory. No black box.
- **Response Planner** — enter your available support; Hosi shows where it saves the most people, driest and least-served first.

*In English,
chiShona,
isiNdebele.*

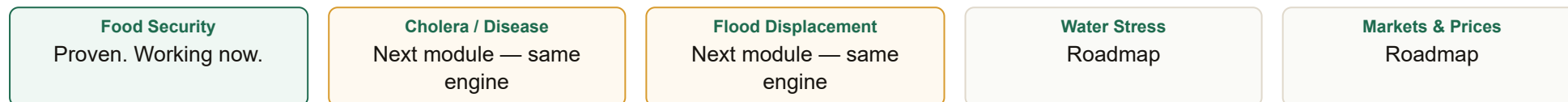
WhatsApp and SMS alerts delivered to district officers on **2G** — no app install required.

THE PLATFORM PLAY

One engine. Many decisions.



The pattern is always the same — build it once, apply it to every national risk:



Prediction is a crowded field. **Prescription is not.** No global tool helps Zimbabwe's own planners decide where to act first — Hosi does.

HOW AI IS USED

We keep AI out of the math. We put it into the language.

LAYER 1 — NUMBERS

Deterministic engine

Transparent, auditable, weighted risk model. Trained on **232 real IPC × CHIRPS observations** (58 districts, 2019–2020). Leave-one-district-out validation.

Every score is traceable arithmetic. No black box. No GPU required.

OLS · $r = 0.484$ · MAE 8.5pp vs 10.2pp global mean

LAYER 2 — REASONING

Claude API (read-only)

Receives only pre-computed numbers from the engine. Writes plain-language early-warning briefs in English, chiShona & isiNdebele. Answers planners' natural-language questions.

Parses incoming field officer reports — turns messy text into clean signals.

Advisory only · never generates data · never touches the model

"The AI never touches the data. The numbers layer is pure deterministic math — so a Minister can audit every score."

VALIDATION — WE PUBLISH OUR BASELINES

Ask other teams for theirs.

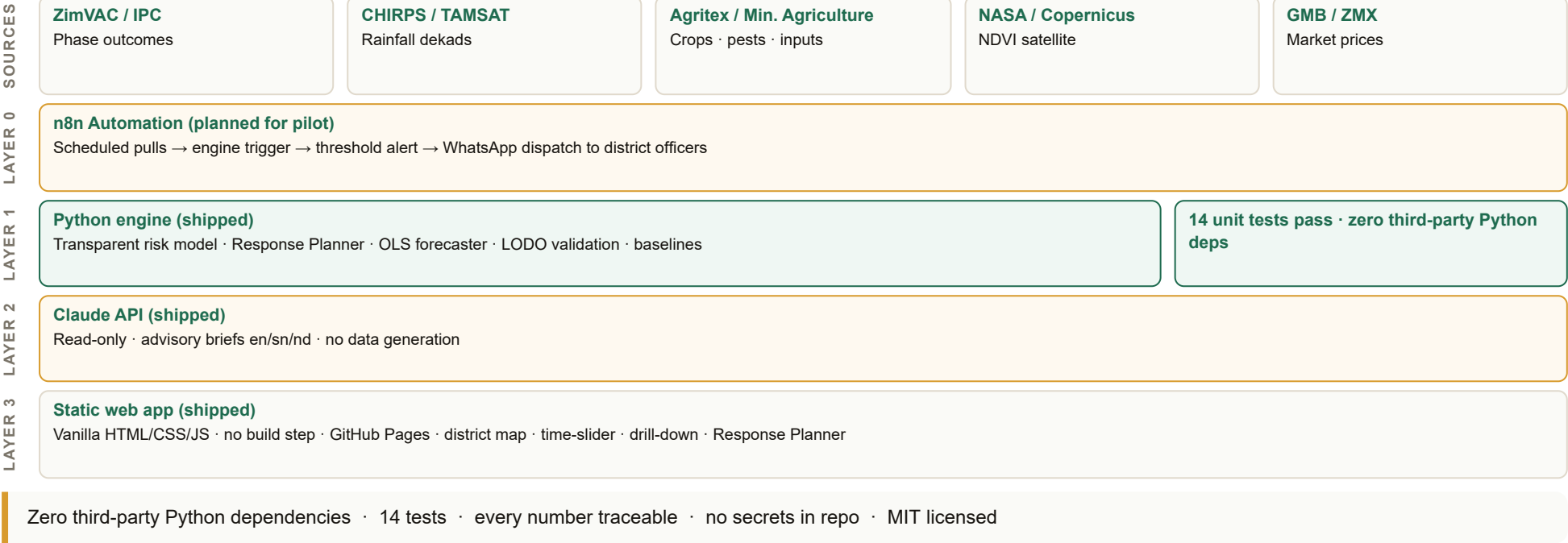
Leave-one-district-out validation · 232 real IPC × CHIRPS observations · 58 districts · 2019–2020

METHOD	R (PEARSON)	MAE (PP)	NOTES
Global-mean baseline	-0.557	10.2	Naive: always predict the panel mean
Persistence baseline	0.129	11.0	Predict last season's outcome
Rainfall-rule baseline	-0.179	10.7	Threshold on CHIRPS rainfall only
OLS model (Hoji)	0.484	8.5	Beats all baselines · vegetation + market features next

$r = 0.81$ is our *internal consistency check* (synthetic data). The table above is our **real validation** — on measured IPC outcomes the model never saw during training. We say what each number is and what it is not.

TECHNICAL ARCHITECTURE

Four layers. Every number traceable.



BUSINESS MODEL

Lean. Open-core. Built to last without a big team.

Pilot monthly operating cost — 20 districts

Item	US\$/mo
VPS — n8n + Python engine (2 GB)	20–40
GitHub Pages hosting	0
Claude API (playbook generation)	30–80
WhatsApp/SMS alert delivery	10–30
Domain + automated backups	5
Total	65–155

- **Phase 1 (0–12 mo):** Donor-funded pilot. Development partner covers operating costs; government operates and owns the data.
 - **Phase 2 (12+ mo):** Per-domain institutional licensing. Each new module (health, climate/disaster) is a new institutional customer — same infrastructure.
 - **Long-term:** Government line-item + ZCHPC national hosting. Operating cost moves off grants onto the National AI Strategy budget.
- Natural pilot funders: WFP Zimbabwe, FAO country office, UNDP. No partnership is claimed — these are the logical candidates.

DEPLOYMENT PLAN

Matabeleland South first. Then everywhere.

Pilot cluster

- **Districts:** Beitbridge · Gwanda · Mangwe (Matabeleland South)
- **Rationale:** Consistently highest food-insecurity risk in ZimVAC data · active ZimVAC and Agritex field presence · isiNdebele alignment
- **Users:** 10–20 district officers and ZimVAC/Agritex staff
- **Delivery:** WhatsApp/SMS alerts on 2G · no app install · static web app
- **Automation:** n8n scheduled CHIRPS pull → engine run → alert dispatch

Milestones

Day 30: MOU signed · demo URL live · compliance complete

Day 60: Live pilot running · alerts delivered · officer feedback logged

Day 90: Warning lead-time measured · adoption case study drafted · Fork A real-label model published · government handover runbook ready

n8n automation keeps the pipeline running with minimal human intervention. Clone repo → install → run. Full restore under 2 hours.

TEAM & THE ASK

Two Zimbabweans. AI-assisted. Fully disclosed.

Nash Barara

PRODUCT & DESIGN LEAD

11-year advertising career · Blender, Adobe Suite, AI creative tools · product vision, UX, pitch, deployment design · Curious Inq founder

Chipo

STRATEGY & COPY LEAD

Strategic framing · stakeholder narrative · policy alignment · ZimVAC & sector context · language copy (en/sn/nd)

WHAT WE ARE ASKING FOR

- Incubation support and structured mentorship
- Introduction to pilot MOU with ZimVAC or Ministry of Agriculture
- Technical mentorship — live data feed integration and ZCHPC hosting pathway

AI build disclosure: Claude and Gemini were used throughout — engine validation, brief generation, document drafting. The architecture decisions, domain framing, and validation methodology are human-owned. We consider this the honest model for AI-assisted teams in 2026.

Prediction is a crowded field. **Prescription is not.**

Demo · [Live URL — to be published on GitHub Pages]

Repo · [github.com/\[handle\]/hozi](https://github.com/[handle]/hozi)

Proposal PDF · [HOZI-TBD_AI4I_Proposal_Development.pdf](#) (see [/proposal/ai4i-2026/](#))

Tatenda. Siyabonga. Thank you.